

# JONATHAN KUNIARDI

SOFTWARE ENGINEERING INTERN — AI & PYTHON

North Jakarta, Indonesia | kuniardijonathan@gmail.com  
linkedin.com/in/jonathankuniardi | fjgb22.github.io | github.com/FJGB22

## PROFESSIONAL SUMMARY

Computer Science undergraduate at BINUS University (Intelligent Systems, GPA 3.72/4.00) who builds real-time conversational AI in Python. Architected a voice-driven AI VTuber pipeline (speech recognition, streaming LLM, interruptible text-to-speech) and benchmarked it to locate the true bottleneck, and own the backend of a five-person group ESP32 voice-assistant project. Available for a full-time internship in semesters 6 and 7 through the BINUS Enrichment Program.

## EDUCATION

**Bachelor of Computer Science** | Intelligent Systems | BINUS University, Jakarta Sep 2024 - Present  
GPA 3.72 / 4.00 · Semester 5 · Expected graduation 2028

**Relevant Coursework:** Machine Learning, Natural Language Processing, Artificial Intelligence, Computer Vision, Data Structures, Software Engineering, Embedded Systems; in progress: Speech Recognition, Deep Learning

**SMA Permai** | Natural Sciences (IPA), Jakarta 2021 - 2024

## ORGANIZATIONAL EXPERIENCE

**Binusian Gaming (BG)** | Regional Secretary, Kemangggisan | Jakarta Jan 2026 - Present

- Administer the formal documentation of a region of ~30 committee members and 100+ members per intake — meeting minutes, event proposals, accountability reports (LPJ), and MoU/MoA.
- Took on the secretary workload from late 2025, months ahead of formal appointment, after more than a year as an activist (since Nov 2024).

**BINUS New Student Expo** | Booth Lead, Binusian Gaming | Jakarta 2025 - 2026

- Spearheaded the organization's booth two years running — designed the concept, recruited and briefed a crew of under 10, and reached 150+ incoming students at each expo.

**BINUS Student Mentoring** | Academic Mentor | Jakarta Sep 2025 - Jan 2026

- Guided the 2 consistently active mentees (of 5 assigned) through the full semester while in my own third semester, explaining material on demand across most of their courses instead of a fixed syllabus.

## SKILLS

**Core:** Python (primary); Java, TypeScript, HTML, CSS (foundational); asyncio & threading; LLM integration (Google Gemini API, OpenAI-compatible clients); prompt & persona design; speech-to-text (faster-whisper); text-to-speech (edge-tts)

**Tools:** Git & GitHub, VS Code, pytest, Ruff, Git pre-commit hooks, Python virtual environments, Figma, Canva, video editing

**Methods:** Modular architecture, performance benchmarking, test-gated commits, AI-assisted development

**Soft Skills:** Technical & formal documentation, event leadership, mentoring

**Spoken Languages:** Indonesian (native), English (intermediate)

## PROJECTS

**Mika — AI VTuber** | Python, asyncio, Gemini API, faster-whisper, edge-tts 2026 - Present

[github.com/FJGB22/ai-vtuber](https://github.com/FJGB22/ai-vtuber) · individual, self-initiated

- Architected a real-time voice pipeline of 6 decoupled modules — speech recognition, turn-taking, streaming LLM, interruptible speech output, avatar control, and configuration — so each stage can be developed and replaced independently.
- Designed sentence-level streaming so speech can start while the model is still generating: replies are split into sentences as tokens arrive and queued for text-to-speech.
- Built a measurement bench (latency, token, jailbreak, and drift tests): over 3 runs, first audio took 4.4 s (median) and speech ~8.6 s, so most of a turn is speech, making interruptible output the priority.
- Measured that the current Gemini endpoint returns each reply in one piece, so sentence streaming saves nothing yet; documented this and the next targets instead of keeping an untested claim.

**Mika IoT — Voice Assistant Satellite** | Python, Google GenAI SDK, faster-whisper, ESP32, pytest, Ruff 2026 - Present

[github.com/FJGB22/mika-voice-assistant](https://github.com/FJGB22/mika-voice-assistant) · group project (5 members), Embedded Systems course

- Own the backend: a Python service that transcribes spoken questions locally (faster-whisper), streams Gemini replies, and keeps multi-turn memory for an Alexa-style ESP32 device, scoped to voice Q&A first.
- Designed a provider-neutral core — history kept in a generic format and translated by one function — so switching LLM vendors means rewriting a single function.
- Enforced code quality with a pre-commit hook that blocks any commit failing lint, format checks, or tests.